Assignment No.2

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Roll No. =100

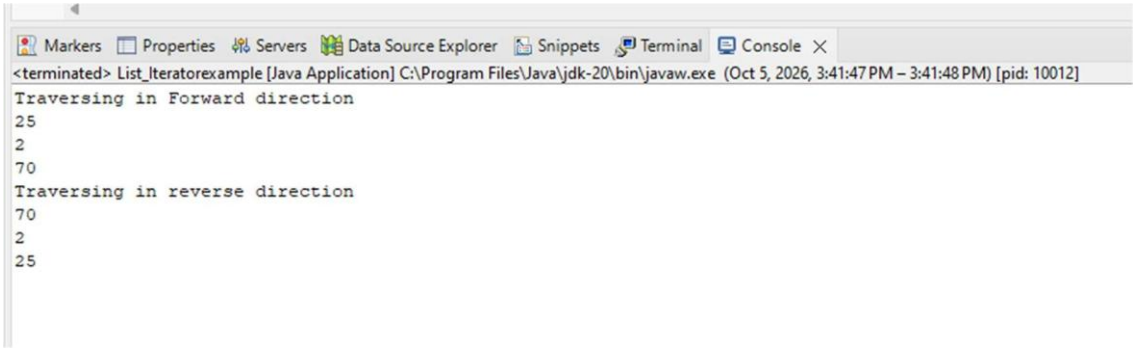
PLO3: Implement Java Collection Framework interfaces for storing, traversing, and manipulating data.

1. **Write a Java program to create List containing list of items and use - ListIterator interface to print items present in the list. Also print the list in reverse/ backward direction.**

   Code:

```
package module2;
import java.util.ArrayList;
import java.util.List;
import java.util.ListIterator;
public class List_Iterator_example {
    public static void main(String[] args) {
        List<Integer> ll = new ArrayList<>();
        ll.add(25);
        ll.add(2);
        ll.add(70);
        ListIterator<Integer> L = ll.listIterator();
        System.out.println("Traversing in Forward direction");
        while (L.hasNext()) {
            System.out.println(L.next());
        }
        System.out.println("Traversing in reverse direction");
        while (L.hasPrevious()) {
            System.out.println(L.previous());
        }
    }
}
```

Output:


Markers
Properties
Servers
Data Source Explorer
Snippets
Terminal
Console
<terminated> List_Iteratorexample [Java Application] C:\Program Files\Java\jdk-20\bin\javaw.exe (Oct 5, 2026, 3:41:47 PM – 3:41:48 PM) [pid: 10012]
Traversing in Forward direction
25
2
70
Traversing in reverse direction
70
2
25

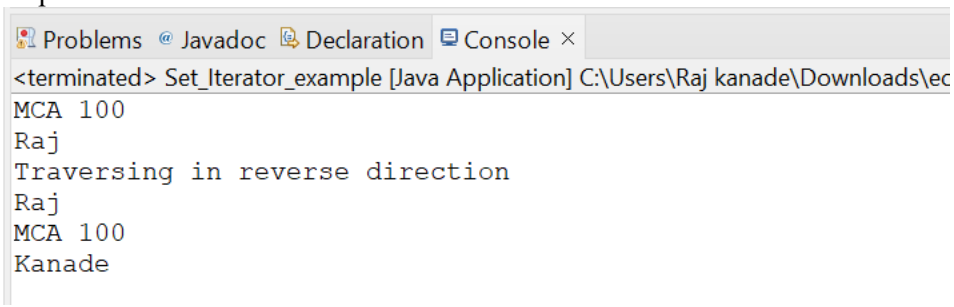
2. **Write a Java program to create a Set containing list of items of type String and print the items in the list using the Iterator interface. Also print the list in reverse/ backward direction.**

Code:

```
package module2;
import java.util.ArrayList;
import java.util.Collections;
import java.util.Iterator;
import java.util.List;
import java.util.ListIterator;
import java.util.Set;
import java.util.TreeSet;
public class Set_Iterator_example {
    public static void main(String[] args) {
        Set<String> s1 = new TreeSet<>();
        s1.add("Aniket");
        s1.add("MCA 70");
        s1.add("Gupta");
        Iterator<String> itr = s1.iterator();
        System.out.println("Traversing in Forward direction");
        while (itr.hasNext()) {
            System.out.println(itr.next());
        }
        System.out.println("Traversing in reverse direction");
        List<String> ll = new ArrayList<String>(s1);
        Collections.reverse(ll);
        for(String language : ll) {
            System.out.println(language);
        }
    }
```

```
}
```

Output:


Problems
Javadoc
Declaration
Console
<terminated> Set_Iterator_example [Java Application] C:\Users\Raj kanade\Downloads\ec
MCA 100
Raj
Traversing in reverse direction
Raj
MCA 100
Kanade

3. **Write a Java program using Set interface containing list of items and perform the following operations:**

   a. Add items in the set.

   b. Insert items of one set into another set.

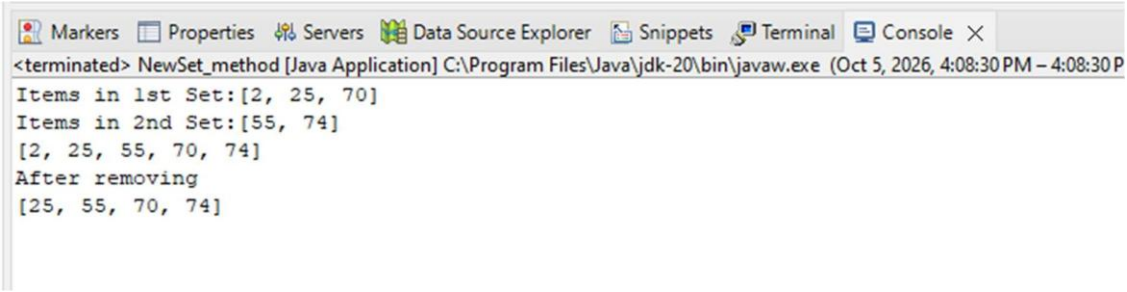
   c. Remove items from the set

   d. Search the specified item in the set

   Code:

```
package module2;
import java.util.Set;
import java.util.TreeSet;
public class NewSet_method {
    public static void main(String[] args) {
        Set<Integer> id = new TreeSet<>();
        id.add(25);
        id.add(2);
        id.add(70);
        System.out.println("Items in 1st Set:" + id);
        Set<Integer> id2 = new TreeSet<>();
        id2.add(55);
        id2.add(74);
        System.out.println("Items in 2nd Set:" + id2);
        id.addAll(id2);
        System.out.println(id);
        if(id.contains(2)) { id.remove(2); }
        System.out.println("After removing");
        System.out.println(id);
    }
}
```

Output:


Markers
Properties
Servers
Data Source Explorer
Snippets
Terminal
Console
<terminated> NewSet_method [Java Application] C:\Program Files\Java\jdk-20\bin\javaw.exe (Oct 5, 2026, 4:08:30 PM – 4:08:30 P
Items in 1st Set:[2, 25, 70]
Items in 2nd Set:[55, 74]
[2, 25, 55, 70, 74]
After removing
[25, 55, 70, 74]

PLO4 : Apply Map and Lambda expressions to perform data operations and functional programming tasks.

1. **Write a Java program using Map interface containing list of items having keys and associated values and perform the following operations:**

   a. Add items in the map.

   b. Remove items from the map

   c. Search specific key from the map

   d. Get value of the specified key

   e. Insert map elements of one map into another map.

   f. Print all keys and values of the map.

   Code:

```
package module2;
import java.util.HashMap;
import java.util.Map;
public class Map_Iterator {
    public static void main(String[] args) {
        Map<Integer,String> map=new HashMap<Integer,String>();
        map.put(1,"Aniket G");
        map.put(2,"Robin S");
        map.put(3,"Rohan G");
        map.forEach((k,v)-> System.out.println(k+ " " +v));

        if(map.containsKey(2)) {
            map.remove(2);
        }
        System.out.println("After removing element with key as 2:");
        map.forEach((k,v)-> System.out.println(k+ " "+v));

        String val = map.get(1);
```

```
        System.out.println("Name of student with roll no 1 is:"+val);
        Map<Integer,String> map2=new HashMap<Integer,String>();
        map2.put(4,"Nikhil G");
        map2.put(5,"Abhishek O");
        map2.put(6,"Hello W");
        map.putAll(map2);
        System.out.println("After Adding one map element into another");
        map.forEach((k,v)-> System.out.println(k+ " "+v));
        for(Integer key : map.keySet()) {
           System.out.println("key: "+key);
        }
        for(String value : map.values()) {
           System.out.println("value: "+value);
        }
    }
}
```

Output:

```
<terminated> Map_Iterator [Java Application] C:\Users\Raj kan
1 Raj K
2 Robin S
3 Rohan G
After removing element with key as 2:
1 Raj K
3 Rohan G
Name of student with roll no 1 is:Raj K
After Adding one map element into another
1 Raj K
3 Rohan G
4 Rahul P
5 Raj O
6 Hello W
key: 1
key: 3
key: 4
key: 5
key: 6
value: Raj K
value: Rohan G
value: Rahul P
value: Raj O
value: Hello W
```

2. **WAP using Lambda Expression with multiple parameters to print addition of two numbers**

Code:

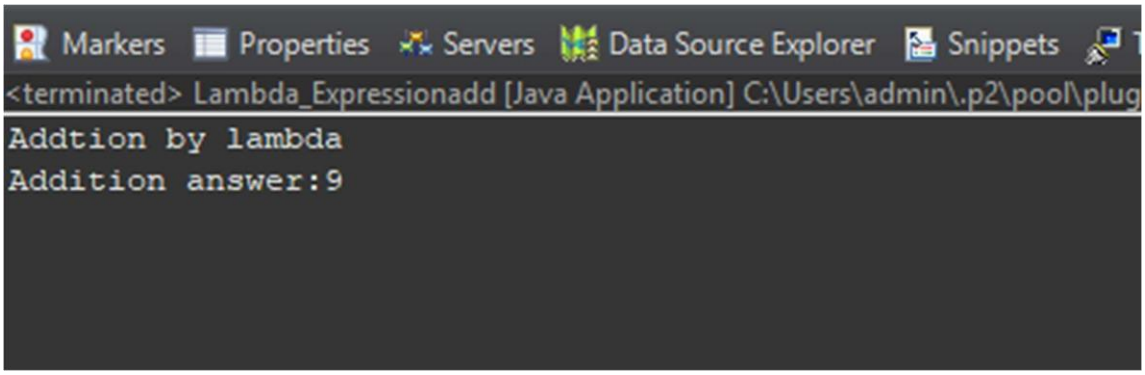
```
package Module_2;


interface Sumoftwono{
	public int Arithmatic(int a,int b);
}


public class Lambda_Expressionadd {


	public static void main(String[] args) {
		System.out.println("Addtion by lambda");
		Sumoftwono why =(a,b)-> a+b;
		int ans;
		ans=why.Arithmatic(4,5);
		System.out.println("Addition answer:" + ans);
	}
}
```

Output:



3. WAP using Lambda Expression to calculate following:

a) Convert Fahrenheit to Celsius

Code:

```
package Module_2;

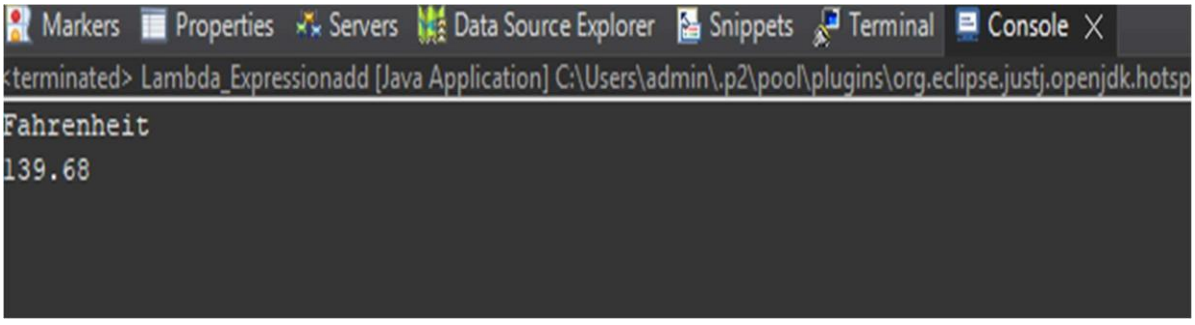

interface Temperaure{
        public void Conversion(double c);
}


public class Lambda_Expressionadd {


        public static void main(String[] args) {
                System.out.println("Fahrenheit ");
                Temperaure why =(c)-> System.out.println((c +32)* 1.8);
                int ans;
                why.Conversion(45.6);
        }


}
```

Output:



b) Convert Kilometers to Meters Code:

```
package Module_2;


interface Distance{
        public void Conversion(double k);
}


public class Lambda_Expressionadd {


        public static void main(String[] args) { System.out.println("KiloMeter
                ");
                Distance why =(k)-> System.out.println(k *1000);
                why.Conversion(6.3);
        }


}
```

Output:

